

Outliers and influential points

Agenda

- + HW 2 due next Friday
- + Today:
 - + More on prediction
 - + Outliers and influential points

Sprinting performance

Last Updated: Thursday, 30 September, 2004, 04:04 GMT 05:04 UK

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Women 'may outsprint men by 2156'

Women sprinters may be outrunning men in the 2156 Olympics if they continue to close the gap at the rate they are doing, according to scientists.

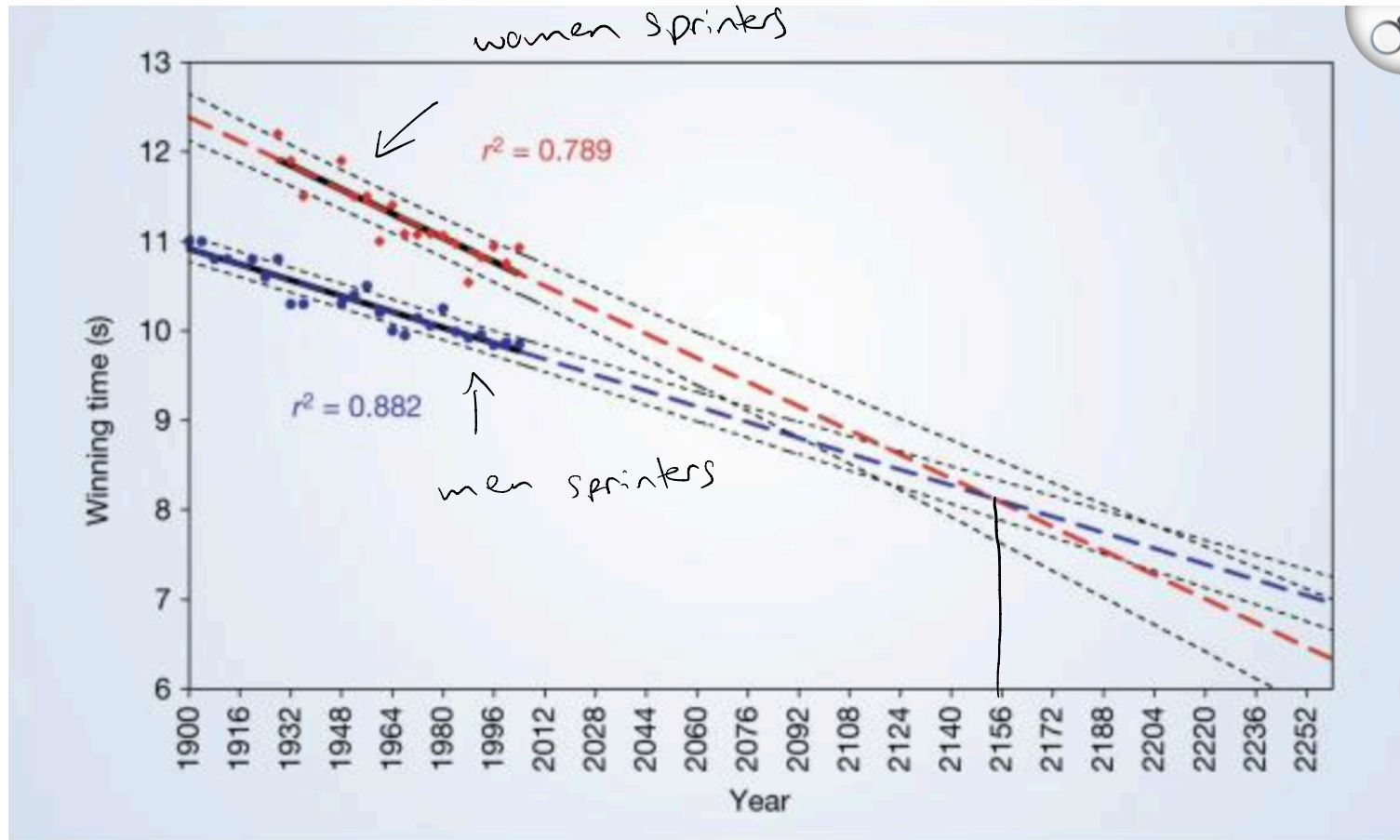
An Oxford University study found that women are running faster than they have ever done over 100m.



Women are set to become the dominant sprinters

At their current rate of improvement, they should overtake men within 150 years, said Dr Andrew Tatem.

Sprinting performance



Extrapolation

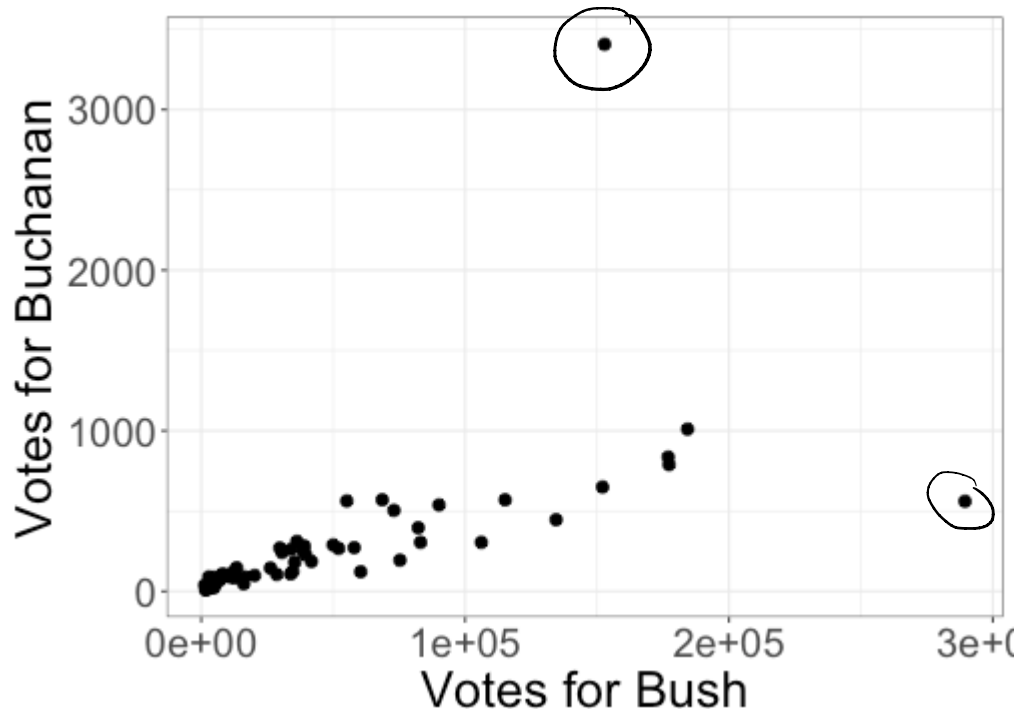
Extrapolation: Making predictions outside the range of observed data

Dangers of extrapolation:

- + Assumes the observed relationship extends to a different population
- + This assumption can lead to nonsensical predictions
- + By definition, we can't examine the relationship outside the range of observed data!

Anything unusual?

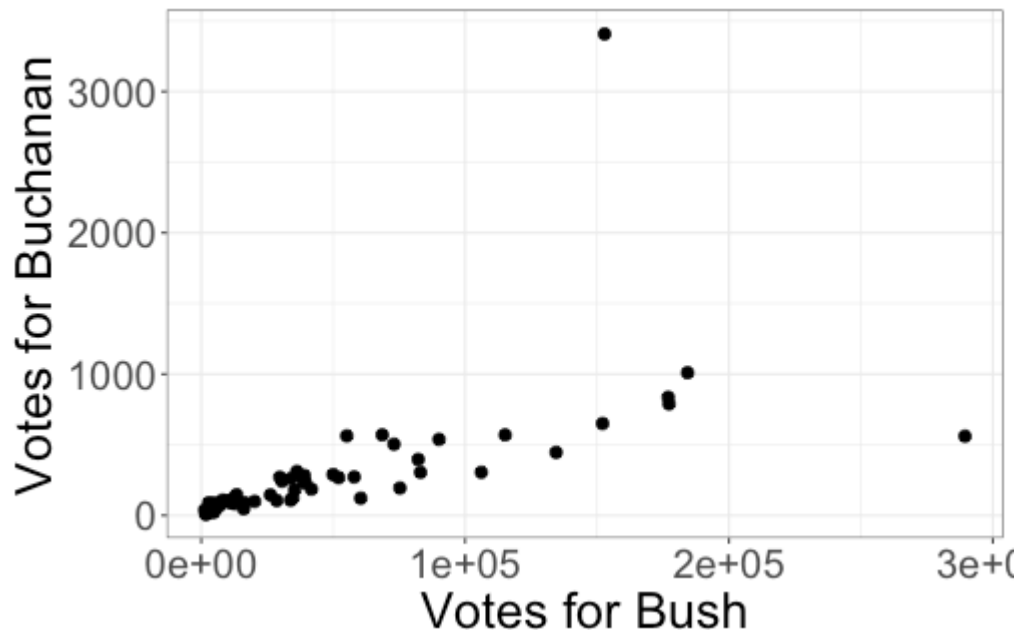
Votes for Bush and Buchanan, in Florida counties in 2000:



Does anything look unusual in this scatterplot?

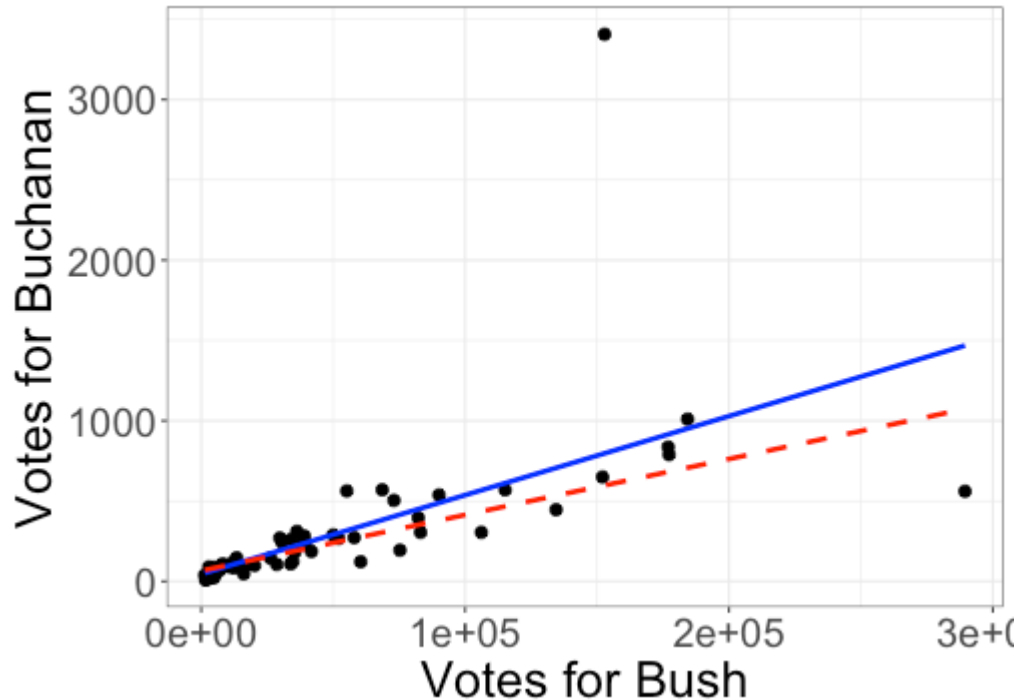
Anything unusual?

Outlier: A point which doesn't follow the general relationship between the predictor and response.



Why might the potential outlier be a problem?

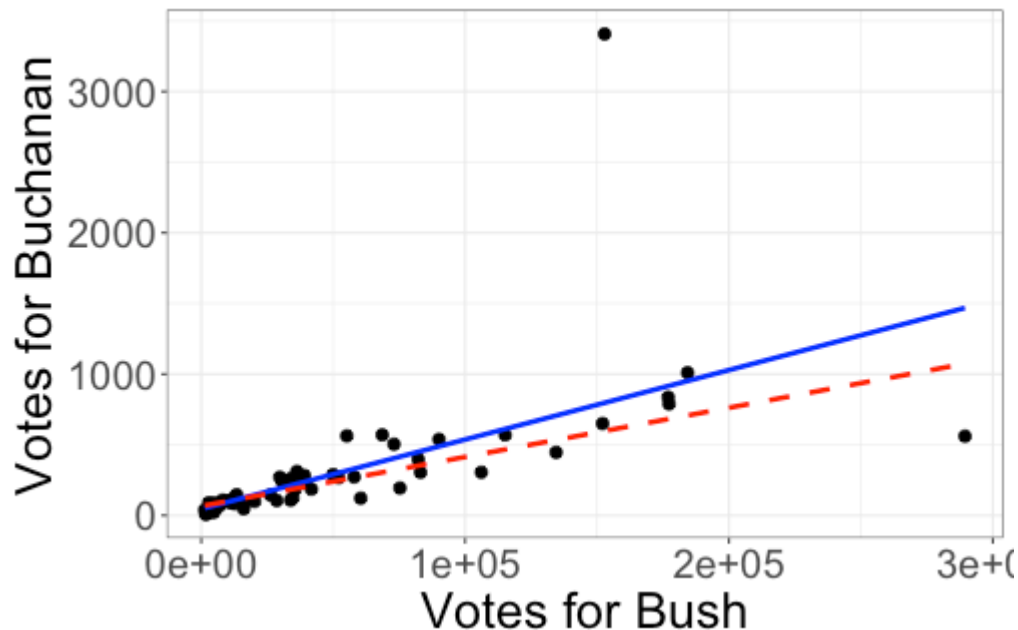
The impact of outliers



- + Blue, solid line: potential outlier is included in the data
- + Red, dashed line: potential outlier is excluded

Influential points

Outliers which substantially change the fitted line are called influential.



Why would an outlier change the fitted line?

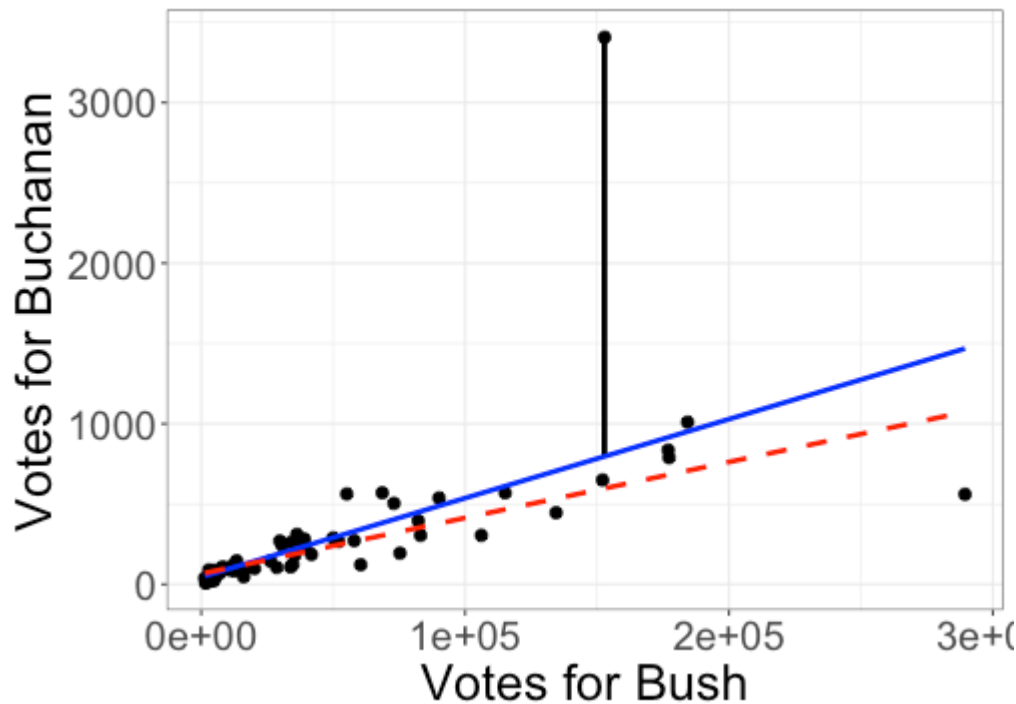
Impact of influential points

SSE

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

$$\Rightarrow y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$$

$$\hat{\beta}_0 \text{ and } \hat{\beta}_1 \text{ minimize } \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$



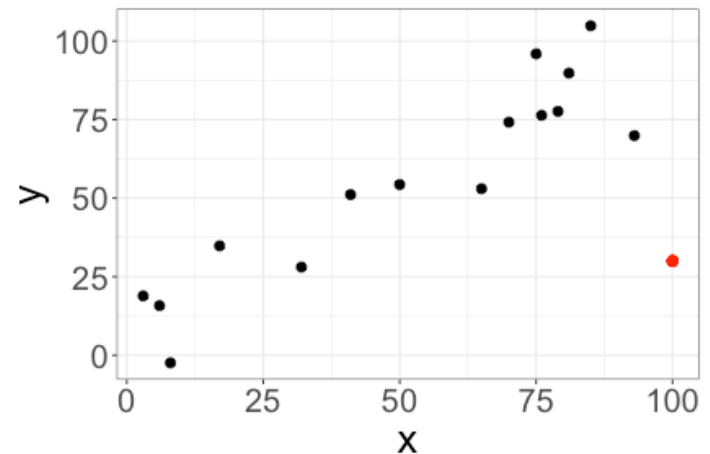
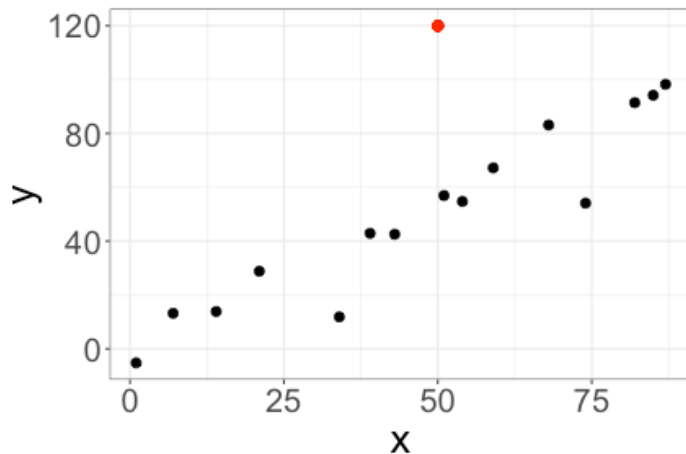
SSE : $\sum_{i=1}^n (y_i - \hat{y}_i)^2$

sum all up $\uparrow$

$\underbrace{(y_i - \hat{y}_i)}_{\text{residuals}}$
 $\underbrace{\quad\quad\quad}_{\text{sum of squared residuals}}$

Potential for influence

Which point do you think is more influential?



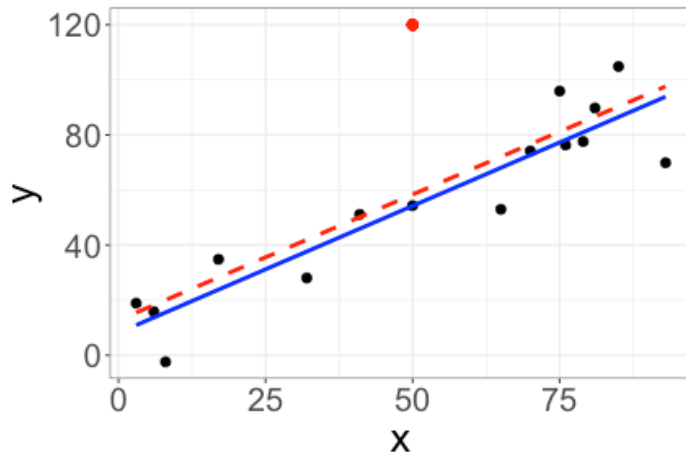
(A) The one on the left

(B) The one on the right

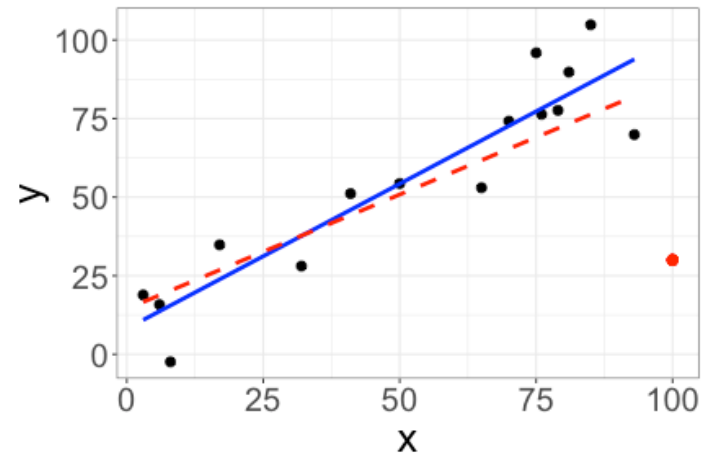
Potential for influence

Influence depends on how far x is from $\bar{x}$.

Some influence



More influence

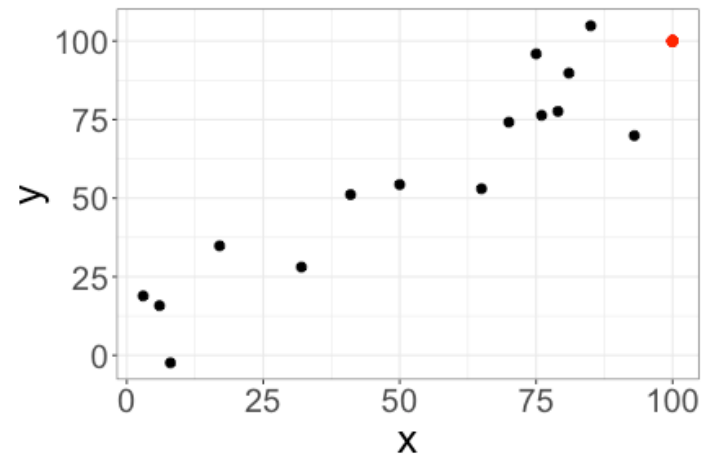
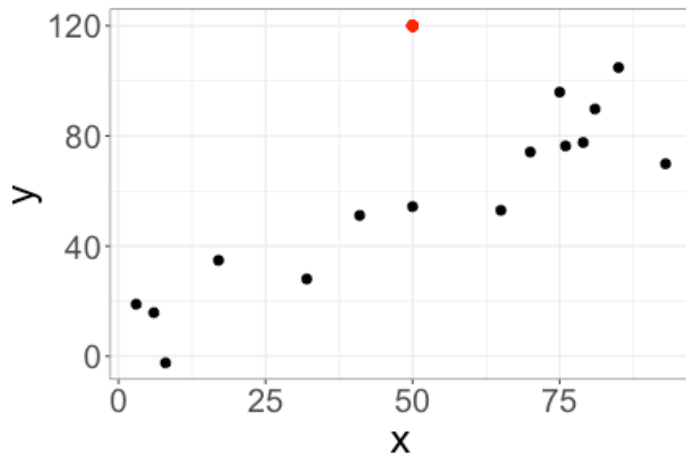


red line: fit with potential outlier

blue line: fit without potential outlier

Influential points

Which point do you think is more influential?



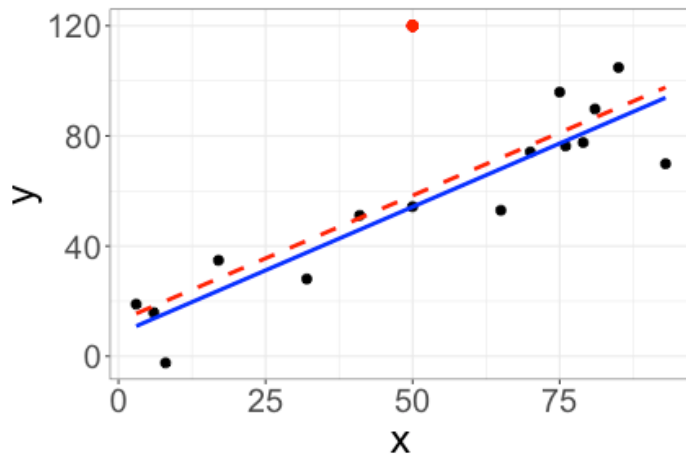
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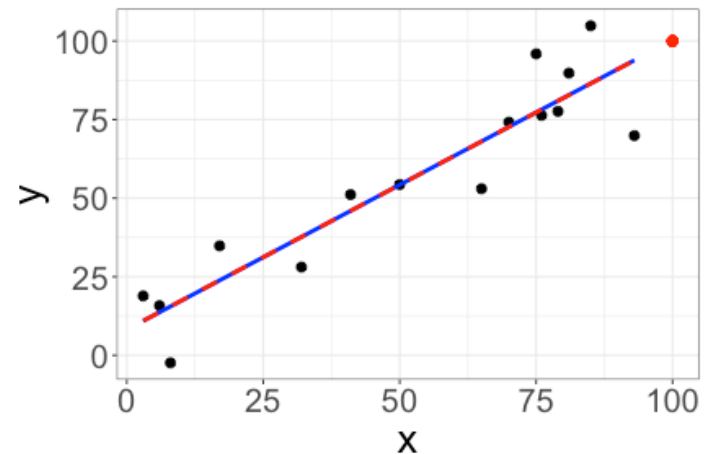
Influential points

Influence depends on both x and y .

Some influence



Not influential



red: line with red point

blue: line without red point

Assessing outliers: studentized residuals

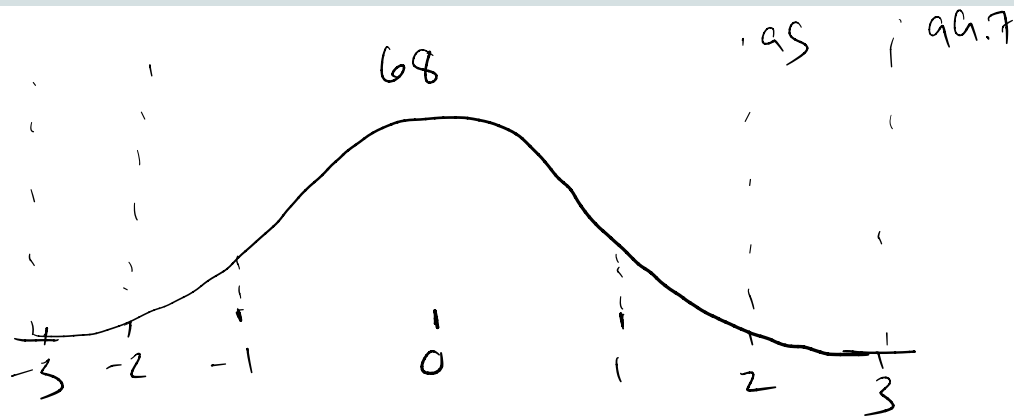
Studentized residuals: $\frac{y_i - \hat{y}_i}{\hat{\sigma}_\varepsilon}$

$y_i - \hat{y}_i$ ← residual
 $\hat{\sigma}_\varepsilon$ ← standard deviation of all residuals (except i -th observation)

where $\hat{\sigma}_\varepsilon$ is computed without observation i .

- ✚ Measure how many *standard deviations* away the observation is from the predicted value

What is a large number of standard deviations?

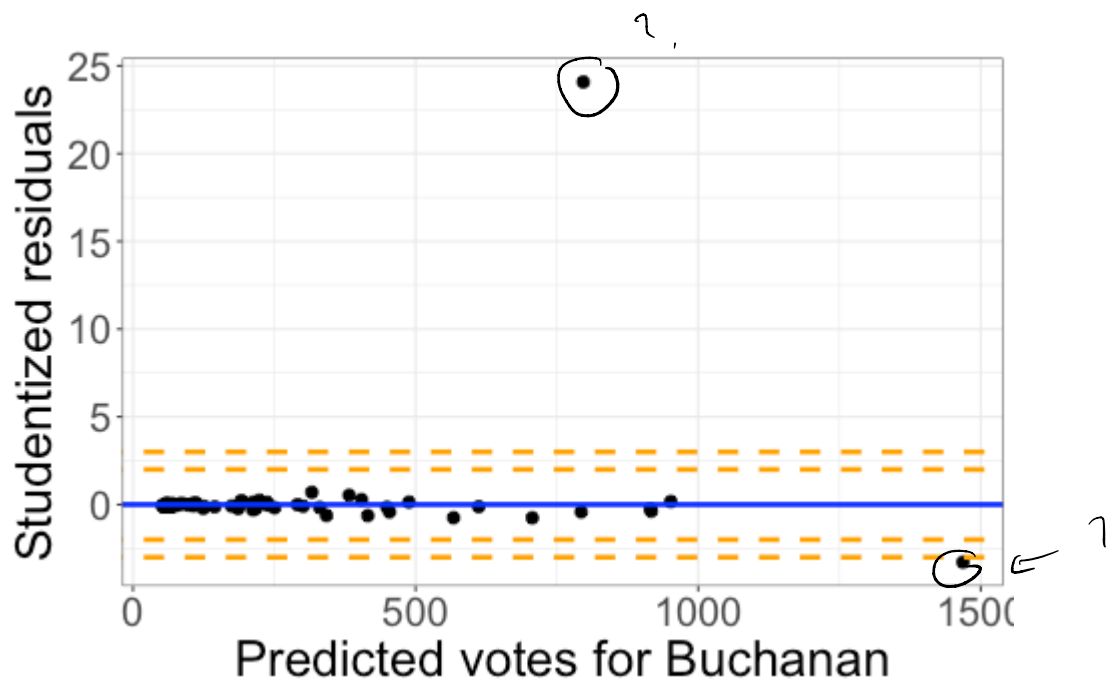


68 - 95 - 99.7 rule
In a normal population,
68% obs. are within $\pm 1SD$
95% within $\pm 2SD$
99.7% within $\pm 3SD$

Studentized residual plot

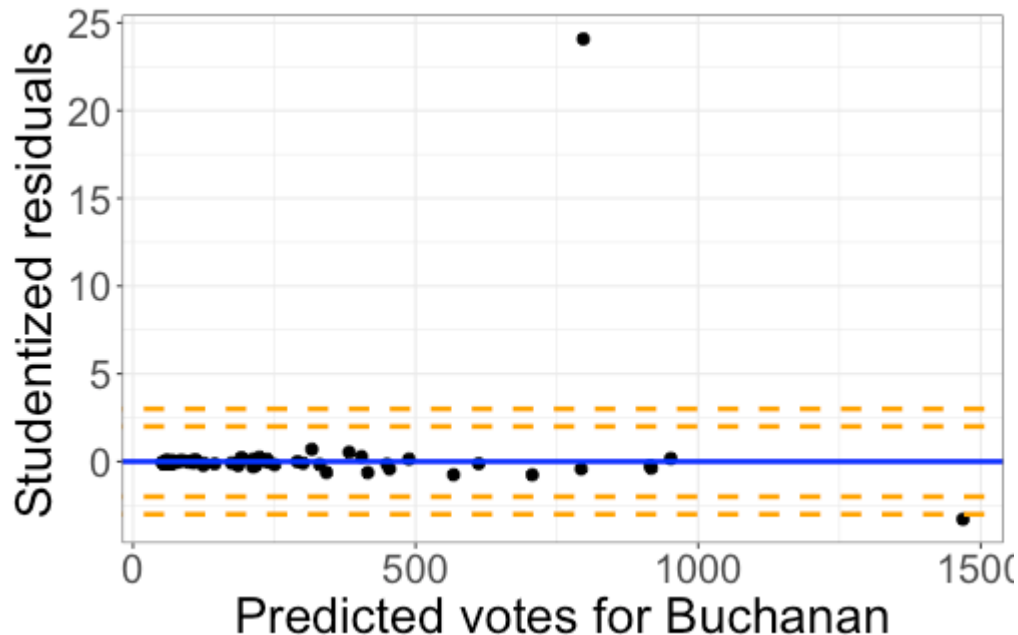
Studentized residuals: $\frac{y_i - \hat{y}_i}{\hat{\sigma}_\varepsilon}$

Studentized residual plot: plot studentized residuals on vertical axis, and $\hat{y}$ on horizontal axis. Add lines at -3, -2, 2, and 3.



Studentized residual plot

Studentized residual plot: plot studentized residuals on vertical axis, and $\hat{y}$ on horizontal axis. Add lines at -3, -2, 2, and 3.



Points < -3 or > 3 are suspected outliers.

Handling outliers

Suppose we identify a suspected outlier. What should we do about it?

Handling outliers

Suppose we identify a suspected outlier. What should we do about it?

If the observation is clearly a measurement error:

- + Remove it

Otherwise:

- + Fit line with and without point, see how the estimated line changes
- + If the estimated line changes substantially, report both fits